Factoring polynomials

1 Consider the expression $m^2 - 16 + 2m + 8$.

a Factor $m^2 - 16$.

b Factor $2m + 8$.

c Using parts (a) and (b) or otherwise, factor $m^2 - 16 + 2m + 8$.

2 Factor:

a $x^4 - 1$ **b** $1 - n^4$ **c** $18 - 8t^2$ **d** $27t^3 - 48t$

e $n^3 - 121n$ **f** $9x^2y^2 - 16z^2w^2$ **g** $16t^2 - 36t^4$ **h** $5t^2 - 20t^4$

i $4p^2q^2 - 81c^2d^2$ **j** $x^2y^2z^2 - 36x^2$ **k** $810p^2 - 10q^2$

3 Factor:

a $pqr - p^2q^2r^2 - p^3q^3r^3$ **b** $-7(-3x + 9) + 8x(-3x + 9)$

c $8x(10y - z + w) - 9u(10y - z + w)$ **d** $5y(y - 10) + 4(10 - y)$.

e $8p(p^2 - 100) - 5(p^2 - 100)$

4 Factor:

a $x^2 - 6x + 9 - y^2$ **b** $x^2 + 8x + 16 - x^4$

c $x^4 - x^2 - 16x - 64$ **d** $9x^2 - 24x + 16 - 4y^2$

e $u^2 - x^2 + 8xy - 16y^2$ **f** $56xy + wx + 56yz + wz$

g $4x + 45yz + 36xy + 5z$ **h** $x^2 - x^2y + xy - x$

i $x^2 - y^2 - x - y$ **j** $m^3 - 9m - 5m^2 + 45$

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5 Factor:

a $5x^2 + 30x + 40$

b $6x^2 - 36x + 48$

c $3x^2 - 12x - 36$

d $w^4 - 5w^2 - 36$

e $20x^2 - 25x - 30$

f $60x^2 - 39x + 6$

g $20x^3 + 25x^2 - 30x$

h $5x^3 + 13x^2 + 6x$

i $80x^4 + 92x^3 + 24x^2$

j $15x^2y + 50xy - 40y$

k $-12x^2y^4 - 57xy^4 + 15y^4$

l $5x^6 - 17x^3 - 12$

m $9p^2 + 24pq + 16q^2$

n $9p^2 - 30pq + 25q^2$

o $p^3 + 8p^2q + 16pq^2$

p $x^4 - 10x^3 + 24x^2$

q $-2x^4 + 2x^3 + 24x^2$

6 Simplify:

a $\dfrac{15k}{20k}$

b $\dfrac{42xy}{18y}$

c $\dfrac{18p^7}{9p^3}$

d $\dfrac{30mn^2}{9m^2n}$

7 Simplify:

a $\dfrac{z+10}{10}$

b $\dfrac{5x^3+4x^5}{x}$

c $\dfrac{18x^3-12x^2+24}{3}$

d $\dfrac{20x^4-35x^3+15x^2+30x}{5x}$

8 Simplify:

a $\dfrac{6x-16}{12}$

b $\dfrac{x-4}{4-x}$

c $\dfrac{3y-45}{15-y}$

d $\dfrac{m^2-100}{m-10}$

e $\dfrac{m^2-1}{1+m}$

f $\dfrac{a^2-81}{9-a}$

g $\dfrac{x^2+4x+4}{10\,(x+2)}$

h $\dfrac{x^2+8x+16}{(x-5)\,(x+4)}$

i $\dfrac{x^2+7x+10}{x+2}$

j $\dfrac{2x^2+4x-198}{5\,(x-9)}$

k $\dfrac{2x+8}{x^2+4x}$

l $\dfrac{20-5x}{2x^2-8x}$

m $\dfrac{m^2-1}{2m+2}$

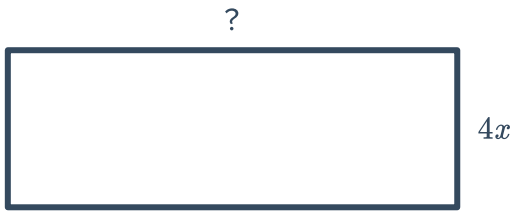
n $\dfrac{2x^2+10x-100}{2x+20}$

o $\dfrac{8x^2+2x-15}{(4x-5)^2}$

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Applications

9 The area of the given rectangle is $\left(4x^4 - 8x\right)$ square units. Find an expression for its length.



10 The area of the given rectangle is $\left(4x^4 - 8x\right)$ square units. Find an expression for its length.



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